Compiler Construction Lab - Assignment 13

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Batch: B3

Objective

To design and implement a spam filter that detects potentially spam messages by matching their contents against predefined Regular Expressions (REs).

Theory

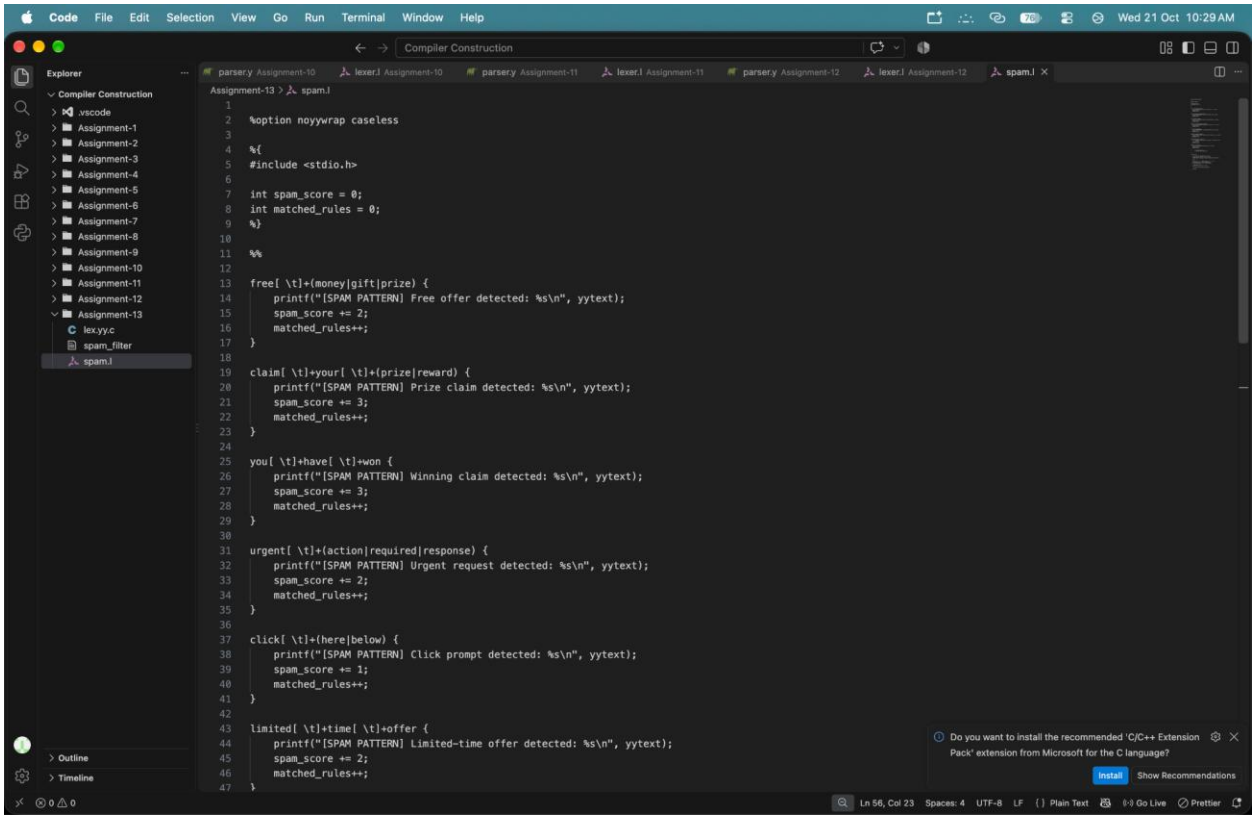
A spam filter is a program that identifies unwanted or suspicious messages, such as promotional messages, fraudulent offers, phishing attempts, and unsolicited advertisements. One simple approach is to use Regular Expressions (REs) to search for specific words, phrases, and patterns that commonly occur in spam messages.

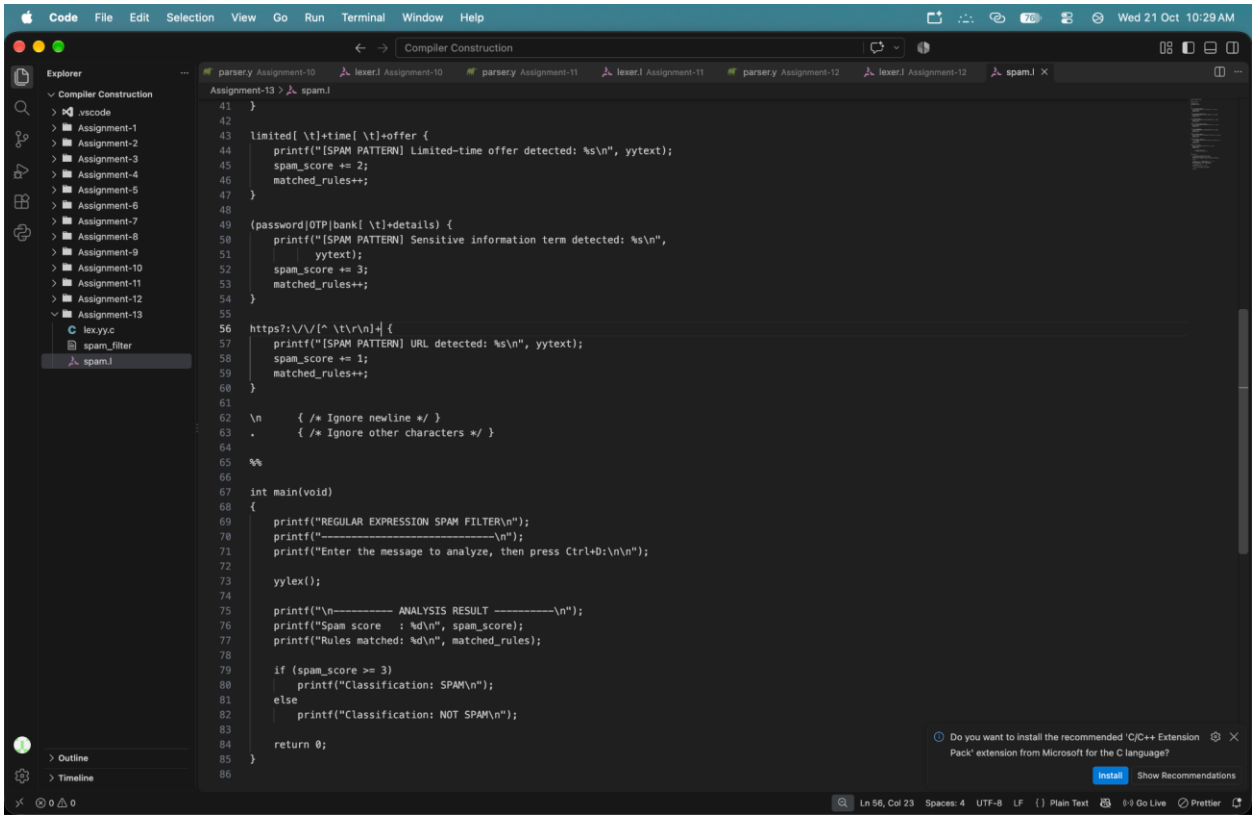
Regular Expressions describe character patterns that can be matched against text. For example, patterns can detect phrases such as “free money,” “claim your prize,” or “urgent action required,” as well as suspicious URLs and requests for sensitive information.

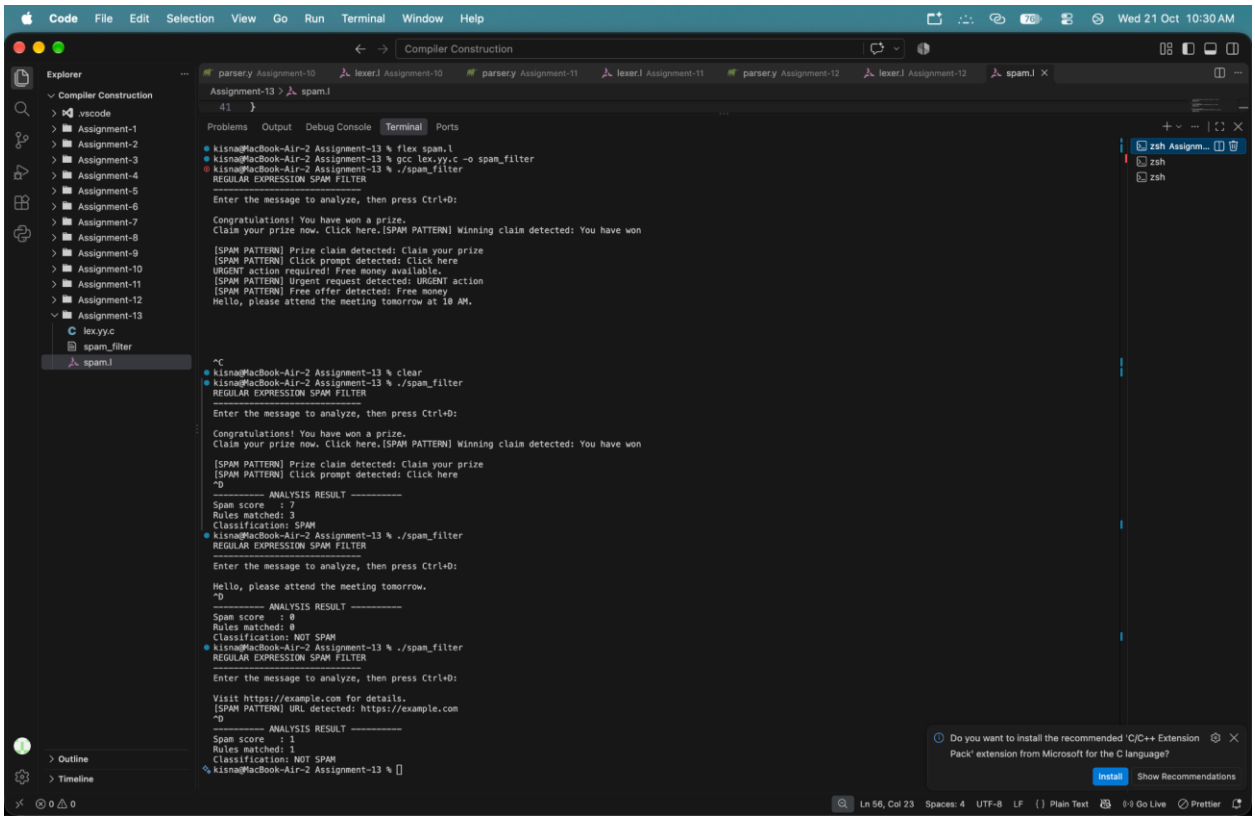
In this assignment, Flex is used to implement a rule-based spam filter. The input message is scanned using predefined regular-expression rules. Each matched pattern contributes to a spam score. Patterns associated with suspicious content receive higher scores, while ordinary text is ignored. After scanning the complete message, the program classifies it as Spam or Not Spam based on a predefined threshold.

This approach is simple, fast, and easy to implement. However, it is rule-based rather than intelligent: legitimate messages can contain suspicious phrases, and sophisticated spam messages may avoid the defined patterns.

Screenshots and Execution Results




Code File Edit Selection View Go Run Terminal Window Help
Wed 21 Oct 10:29 AM
Compiler Construction
Explorer
Compiler Construction
.vscode
Assignment-1
Assignment-2
Assignment-3
Assignment-4
Assignment-5
Assignment-6
Assignment-7
Assignment-8
Assignment-9
Assignment-10
Assignment-11
Assignment-12
Assignment-13
lex.yy.c
spam_filter
spam.l
parser.y Assignment-10
lexer.l Assignment-10
parser.y Assignment-11
lexer.l Assignment-11
parser.y Assignment-12
lexer.l Assignment-12
spam.l
Assignment-13 > spam.l
41 }
42
43 limited[\t]+time[\t]+offer {
44 printf("[SPAM PATTERN] Limited-time offer detected: %s\n", yytext);
45 spam_score += 2;
46 matched_rules++;
47 }
48
49 (password|OTP|bank[\t]+details) {
50 printf("[SPAM PATTERN] Sensitive information term detected: %s\n",
51 yytext);
52 spam_score += 3;
53 matched_rules++;
54 }
55
56 https?:\/\/[^ \t\r\n]+ {
57 printf("[SPAM PATTERN] URL detected: %s\n", yytext);
58 spam_score += 1;
59 matched_rules++;
60 }
61
62 \n { /* Ignore newline */ }
63 . { /* Ignore other characters */ }
64
65 %%
66
67 int main(void)
68 {
69 printf("REGULAR EXPRESSION SPAM FILTER\n");
70 printf("------------------------------\n");
71 printf("Enter the message to analyze, then press Ctrl+D:\n\n");
72
73 yylex();
74
75 printf("\n---------- ANALYSIS RESULT ----------\n");
76 printf("Spam score : %d\n", spam_score);
77 printf("Rules matched: %d\n", matched_rules);
78
79 if (spam_score >= 3)
80 printf("Classification: SPAM\n");
81 else
82 printf("Classification: NOT SPAM\n");
83
84 return 0;
85 }
86
Outline
Timeline
Do you want to install the recommended 'C/C++ Extension Pack' extension from Microsoft for the C language?
Install
Show Recommendations
Ln 56, Col 23 Spaces: 4 UTF-8 LF { } Plain Text Go Live Prettier


Code File Edit Selection View Go Run Terminal Window Help
Wed 21 Oct 10:30 AM
Compiler Construction
Problems Output Debug Console Terminal Ports
kisna@MacBook-Air-2 Assignment-13 % flex spam.l
kisna@MacBook-Air-2 Assignment-13 % gcc lex.yy.c -o spam_filter
kisna@MacBook-Air-2 Assignment-13 % ./spam_filter
REGULAR EXPRESSION SPAM FILTER
Enter the message to analyze, then press Ctrl+D:
Congratulations! You have won a prize.
Claim your prize now. Click here.[SPAM PATTERN] Winning claim detected: You have won
[SPAM PATTERN] Prize claim detected: Claim your prize
[SPAM PATTERN] Click prompt detected: Click here
URGENT action required! Free money available.
[SPAM PATTERN] Urgent request detected: URGENT action
[SPAM PATTERN] Free offer detected: Free money
Hello, please attend the meeting tomorrow at 10 AM.
^C
kisna@MacBook-Air-2 Assignment-13 % clear
kisna@MacBook-Air-2 Assignment-13 % ./spam_filter
REGULAR EXPRESSION SPAM FILTER
Enter the message to analyze, then press Ctrl+D:
Congratulations! You have won a prize.
Claim your prize now. Click here.[SPAM PATTERN] Winning claim detected: You have won
[SPAM PATTERN] Prize claim detected: Claim your prize
[SPAM PATTERN] Click prompt detected: Click here
^D
---------- ANALYSIS RESULT ----------
Spam score : 7
Rules matched: 3
Classification: SPAM
kisna@MacBook-Air-2 Assignment-13 % ./spam_filter
REGULAR EXPRESSION SPAM FILTER
Enter the message to analyze, then press Ctrl+D:
Hello, please attend the meeting tomorrow.
^D
---------- ANALYSIS RESULT ----------
Spam score : 0
Rules matched: 0
Classification: NOT SPAM
kisna@MacBook-Air-2 Assignment-13 % ./spam_filter
REGULAR EXPRESSION SPAM FILTER
Enter the message to analyze, then press Ctrl+D:
Visit https://example.com for details.
[SPAM PATTERN] URL detected: https://example.com
^D
---------- ANALYSIS RESULT ----------
Spam score : 1
Rules matched: 1
Classification: NOT SPAM
kisna@MacBook-Air-2 Assignment-13 %
zsh Assignm...
zsh
zsh
Do you want to install the recommended 'C/C++ Extension Pack' extension from Microsoft for the C language?
Install
Show Recommendations
Ln 56, Col 23 Spaces: 4 UTF-8 LF { } Plain Text Go Live Prettier